

# Krydo

Privacy-Preserving Financial Trust Infrastructure

## Business Plan & Technical Strategy

Stellar Build Station Delhi | 21-Day Residency

Powered by Rise In  
stellar.org | risein.com

---

|                    |                                        |
|--------------------|----------------------------------------|
| <b>Program</b>     | Stellar Build Station Delhi            |
| <b>Organisers</b>  | Stellar & Rise In                      |
| <b>Location</b>    | Innov8 CP, Delhi NCR                   |
| <b>Duration</b>    | 21 Days (4 July – 25 July 2026)        |
| <b>Cohort</b>      | 15 Selected Teams                      |
| <b>Format</b>      | Hybrid with Weekly In-Person Check-ins |
| <b>Build Hours</b> | 11:00 AM – 4:30 PM                     |

---

*Prove you qualify — without revealing what you have.*

Confidential Business Document | July 2026

## Contents

---

|          |                                                          |           |
|----------|----------------------------------------------------------|-----------|
| <b>1</b> | <b>Executive Summary</b>                                 | <b>2</b>  |
| <b>2</b> | <b>The Problem</b>                                       | <b>2</b>  |
| 2.1      | Over-Collection Is the Default . . . . .                 | 2         |
| 2.2      | Why Existing Solutions Fall Short . . . . .              | 3         |
| <b>3</b> | <b>Product Overview: Krydo</b>                           | <b>3</b>  |
| 3.1      | Value Proposition . . . . .                              | 3         |
| 3.2      | Supported Proof Types . . . . .                          | 3         |
| 3.3      | End-to-End Flow (Income Verification Example) . . . . .  | 3         |
| 3.4      | Current Technical Stack (Ethereum MVP) . . . . .         | 4         |
| <b>4</b> | <b>Why Stellar? Strategic Rationale</b>                  | <b>4</b>  |
| 4.1      | Network Fit for Krydo . . . . .                          | 4         |
| 4.2      | Stellar vs. Ethereum for Production . . . . .            | 5         |
| 4.3      | Why Now: Stellar Garage Delhi . . . . .                  | 5         |
| <b>5</b> | <b>Business Model &amp; Revenue Strategy</b>             | <b>5</b>  |
| 5.1      | Revenue Philosophy . . . . .                             | 5         |
| 5.2      | Primary Revenue Streams . . . . .                        | 5         |
| 5.2.1    | Stream 1: Per-Verification API (Core) . . . . .          | 5         |
| 5.2.2    | Stream 2: SaaS Subscription (Verifiers) . . . . .        | 6         |
| 5.2.3    | Stream 3: Issuer Onboarding & Licensing . . . . .        | 6         |
| 5.2.4    | Stream 4: White-Label & B2B2C . . . . .                  | 6         |
| 5.2.5    | Stream 5: Ecosystem Grants & Strategic Funding . . . . . | 6         |
| 5.3      | Unit Economics . . . . .                                 | 6         |
| 5.4      | Revenue Milestones . . . . .                             | 7         |
| <b>6</b> | <b>Target Market &amp; Use Cases</b>                     | <b>7</b>  |
| 6.1      | Primary Customer Segments . . . . .                      | 7         |
| 6.2      | India-First Use Cases (Priority Order) . . . . .         | 7         |
| 6.3      | Total Addressable Market (India, Directional) . . . . .  | 7         |
| <b>7</b> | <b>Go-to-Market Strategy</b>                             | <b>8</b>  |
| 7.1      | Phase 1: Residency & Proof (July 2026) . . . . .         | 8         |
| 7.2      | Phase 2: Paid Pilot (August–October 2026) . . . . .      | 8         |
| 7.3      | Phase 3: Scale (Q4 2026–2027) . . . . .                  | 8         |
| 7.4      | Sales Motion . . . . .                                   | 8         |
| 7.5      | Key Messaging . . . . .                                  | 8         |
| <b>8</b> | <b>Stellar Garage Delhi: Program Plan</b>                | <b>8</b>  |
| 8.1      | Program Overview . . . . .                               | 9         |
| 8.2      | In-Person Milestone Schedule . . . . .                   | 9         |
| 8.3      | Builder Expectations & Compliance . . . . .              | 10        |
| 8.4      | 21-Day Build Timeline . . . . .                          | 11        |
| <b>9</b> | <b>Technical Roadmap: Ethereum to Stellar</b>            | <b>11</b> |
| 9.1      | Migration Scope . . . . .                                | 11        |
| 9.2      | MVP Scope Constraint (21 Days) . . . . .                 | 12        |
| 9.3      | Architecture on Stellar . . . . .                        | 12        |
| 9.4      | Post-Residency Technical Debt . . . . .                  | 12        |

|                                                    |           |
|----------------------------------------------------|-----------|
| <b>10 Demo Day Strategy (25 July 2026)</b>         | <b>12</b> |
| 10.1 Three-Act Live Demo Script (5–7 Minutes)      | 12        |
| 10.2 Pitch Deck Structure (Recommended Slides)     | 12        |
| 10.3 Backup Plan                                   | 13        |
| <b>11 Financial Projections (Directional)</b>      | <b>13</b> |
| 11.1 Year 1 P&L Summary (Conservative Scenario)    | 13        |
| 11.2 Funding Strategy                              | 13        |
| <b>12 Risk Analysis &amp; Mitigation</b>           | <b>14</b> |
| <b>13 Competitive Moat &amp; Defensibility</b>     | <b>14</b> |
| <b>14 Conclusion &amp; Immediate Next Steps</b>    | <b>14</b> |
| 14.1 Immediate Action Items                        | 15        |
| <b>A Appendix A: API Endpoints (Reference)</b>     | <b>15</b> |
| <b>B Appendix B: Progress Update Template</b>      | <b>15</b> |
| <b>C Appendix C: Document &amp; Resource Links</b> | <b>16</b> |

## 1 Executive Summary

Krydo is a **privacy-preserving verifiable credential platform** that enables users to prove financial and identity predicates—such as “*credit score  $\geq 700$* ” or “*annual income  $\geq \text{INR } 10 \text{ Lakh}$* ”—**without disclosing the underlying values**. Built on zero-knowledge sigma protocols over Pedersen commitments, Krydo addresses a critical gap in India’s rapidly evolving compliance landscape: verification flows that over-collect personal data by default.

This document outlines the business strategy, revenue model, go-to-market plan, and technical roadmap for deploying Krydo on the Stellar network as part of the **Stellar Garage Delhi** 21-day residency program.

### Strategic Thesis

- **Problem:** Lenders, landlords, and fintechs receive full salary slips, bank statements, and PAN data when they only need a yes/no qualification answer.
- **Solution:** Krydo anchors credential trust on-chain; users generate ZK proofs in-browser; verifiers learn predicates, not PII.
- **Why Stellar:** Sub-cent transaction costs, Soroban smart contracts, cross-border settlement, and ecosystem support via Rise In sponsorship for mainnet deployment.
- **Business model:** B2B SaaS and per-verification API fees charged to *verifiers* (NBFCs, fintechs, exchanges) and *issuers* (employers, KYC providers).
- **Near-term goal:** Deliver a mainnet-ready demo by 25 July 2026; secure one paid pilot within 90 days post-residency.

## 2 The Problem

### 2.1 Over-Collection Is the Default

Current identity and financial verification in India follows a destructive pattern. When a lender asks “*Does this applicant earn at least INR 10 Lakh per annum?*”, the typical response involves:

- Full salary slips and employer details
- Six months of bank statements
- PAN, Aadhaar-linked KYC documents
- Credit bureau reports with complete score history

This data is stored indefinitely by multiple parties, re-sold through data brokers, and cannot be revoked once leaked. Under India’s **Digital Personal Data Protection Act (DPDP)**, data minimisation is shifting from best practice to legal requirement.

## 2.2 Why Existing Solutions Fall Short

Table 1: Competitive landscape and Krydo differentiation

| Approach                                          | Limitation                                         | Krydo Advantage                                           |
|---------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------|
| <b>Traditional KYC (Signzy, IDfy, HyperVerge)</b> | Stores full PII; breach liability; poor revocation | Predicate-only verification; on-chain revocation          |
| <b>Manual document upload</b>                     | Slow, fraud-prone, poor UX                         | Cryptographic proof in milliseconds                       |
| <b>SNARK-based ZK systems</b>                     | Trusted setup, circuit compilation, high gas       | Sigma protocols: no setup ceremony, browser-native proofs |
| <b>Self-attested claims</b>                       | No trust anchor                                    | On-chain issuer whitelist + credential anchoring          |

## 3 Product Overview: Krydo

### 3.1 Value Proposition

Krydo operates as a **four-actor trust network**:

1. **Root Authority** — Deploys contracts and whitelists trusted issuers.
2. **Licensed Issuer** — Banks, employers, CIBIL mirrors, KYC providers issue signed credentials.
3. **Holder (User)** — Receives credentials; generates ZK proofs off-chain.
4. **Verifier** — Lenders, exchanges, landlords verify proofs without accessing underlying data.

### 3.2 Supported Proof Types

Table 2: Zero-knowledge proof catalogue (MVP)

| TableHead            |                                         |                         |
|----------------------|-----------------------------------------|-------------------------|
| range_above          | Credit score $\geq 700$                 | Predicate holds         |
| range_below          | Debt ratio $\leq 40\%$                  | Predicate holds         |
| equality             | Resident of India                       | Match, not full address |
| membership           | Citizenship $\in \{\text{IN, US, UK}\}$ | Set membership          |
| non_zero             | Has valid PAN number                    | Existence only          |
| selective_disclosure | Reveal name + employer; hide salary     | Chosen fields only      |

### 3.3 End-to-End Flow (Income Verification Example)

1. **Issuance:** Employer (whitelisted issuer) signs credential; hash anchored on-chain; plaintext stored encrypted off-chain.

2. **Proof generation:** User creates Pedersen commitment  $C = v \cdot G + r \cdot H$  and sigma proof  $\pi$  in the browser (no gas).
3. **Verification:** Verifier submits `proofId` to API; server re-runs EC math; checks issuer still whitelisted on-chain; returns `{ valid: true }`.
4. **Outcome:** Lender confirms  $\text{income} \geq \text{threshold}$ . Actual salary remains hidden.

### 3.4 Current Technical Stack (Ethereum MVP)

| Layer           | Technology                                                      |
|-----------------|-----------------------------------------------------------------|
| Smart contracts | Solidity 0.8.x (KrydoAuthority, KrydoCredentials, KrydoAudit)   |
| Cryptography    | Sigma protocols, Pedersen commitments on secp256k1, Fiat-Shamir |
| Backend         | Node.js 20, Express, TypeScript, Zod, Firebase Firestore        |
| Frontend        | React 18, Vite, shadcn/ui, TanStack Query                       |
| Wallet          | wagmi v2 + RainbowKit (EIP-1193)                                |
| Auth            | EIP-4361 SIWE + JWT                                             |
| Testing         | 154 unit tests (Vitest)                                         |

## 4 Why Stellar? Strategic Rationale

### 4.1 Network Fit for Krydo

Stellar is not merely a deployment target—it is a **strategic enabler** for Krydo’s business model in India and cross-border markets.

#### Stellar Advantages for Krydo

1. **Cost efficiency:** Sub-cent transaction fees enable high-volume verification anchoring without prohibitive gas costs. Ethereum mainnet gas makes per-credential anchoring economically unviable at scale.
2. **Soroban smart contracts:** Rust-based WASM contracts for issuer registry, credential hash anchoring, and audit events—with formal verification tooling and predictable execution costs.
3. **Cross-border credentials:** Stellar’s remittance and payment rails align with verifying income/credit for diaspora lending, global DeFi, and international student financing.
4. **Fast finality:** 3–5 second confirmation supports real-time verification UX expected by fintech customers.
5. **Ecosystem support:** Stellar Garage Delhi, Rise In mentorship, mainnet sponsorship, and SCF grant pathways reduce go-to-market friction.
6. **Regulatory narrative:** Stellar’s focus on financial inclusion and compliant payment infrastructure resonates with NBFC and banking pilot conversations in India.

## 4.2 Stellar vs. Ethereum for Production

Table 3: Deployment platform comparison

| Dimension               | Ethereum Mainnet     | Stellar Mainnet               |
|-------------------------|----------------------|-------------------------------|
| Avg. tx cost            | \$1–\$50+ (variable) | \$0.00001–\$0.001             |
| Contract language       | Solidity (EVM)       | Rust (Soroban)                |
| Finality                | ~12 sec (L1)         | 3–5 sec                       |
| Cross-border focus      | DeFi-native          | Payments & remittance native  |
| India fintech narrative | Speculation concerns | Financial inclusion alignment |
| Residency support       | None                 | Rise In mainnet sponsorship   |

## 4.3 Why Now: Stellar Garage Delhi

The 21-day residency provides a **forced function** for mainnet deployment, structured feedback, and ecosystem introductions that would otherwise take 6–12 months to replicate organically. Program requirements—mainnet live by 25 July 2026—align directly with investor and pilot customer expectations.

# 5 Business Model & Revenue Strategy

## 5.1 Revenue Philosophy

Krydo is **B2B infrastructure**. End users (credential holders) access the platform free of charge during growth phase. Revenue is generated from organisations that *issue* or *verify* credentials at scale.

### Critical Principle

Do **not** monetise holders directly in Phase 1. Friction at the user layer kills adoption. Verifiers and issuers already budget for KYC/compliance spend—redirect that budget to Krydo.

## 5.2 Primary Revenue Streams

### 5.2.1 Stream 1: Per-Verification API (Core)

Charge verifiers for each successful ZK proof verification via REST API.

Table 4: Per-verification pricing tiers

| Verification Type         | Price (INR) | Est. Cost | Gross Margin |
|---------------------------|-------------|-----------|--------------|
| Income range proof        | 20–40       | 1–3       | 85–95%       |
| Credit score predicate    | 25–50       | 1–3       | 85–95%       |
| Identity membership       | 10–25       | 0.5–2     | 85–95%       |
| Full selective disclosure | 40–80       | 2–5       | 85–95%       |

**Example:** A small NBFC processing 10,000 loan applications/month at INR 20/verification

generates **INR 2 Lakh/month** (24 Lakh ARR).

### 5.2.2 Stream 2: SaaS Subscription (Verifiers)

Table 5: Verifier subscription plans

| Plan       | Monthly (INR)   | Includes                                            |
|------------|-----------------|-----------------------------------------------------|
| Starter    | 15,000–25,000   | 1,000 verifications, dashboard, API keys            |
| Growth     | 50,000–1,00,000 | 10,000 verifications, webhooks, SLA, analytics      |
| Enterprise | Custom          | Unlimited volume, dedicated support, custom schemas |

### 5.2.3 Stream 3: Issuer Onboarding & Licensing

- **One-time setup fee:** INR 50,000–5,00,000 (integration, legal review, whitelist activation)
- **Annual issuer license:** INR 1–10 Lakh (maintain whitelisted issuer status on-chain)
- **Per-credential issuance (optional):** INR 2–10 per anchored credential

### 5.2.4 Stream 4: White-Label & B2B2C

- **Setup fee:** INR 5–20 Lakh for embedded “Verify with Krydo” integration
- **Revenue share:** 10–20% of verifier fees on partner-sourced volume
- **Target partners:** Payroll SaaS (Keka, RazorpayX), HR platforms, ed-tech credential issuers

### 5.2.5 Stream 5: Ecosystem Grants & Strategic Funding

- Stellar Community Fund (SCF) grants for infrastructure build and maintenance
- Rise In / Stellar Foundation ecosystem partnerships
- Cross-border pilot funding for remittance-linked verification use cases

## 5.3 Unit Economics

Table 6: Unit economics per verification

| Metric                                  | Value                  |
|-----------------------------------------|------------------------|
| Average revenue per verification (ARPV) | INR 15–30              |
| Variable cost (API, DB, Stellar tx)     | INR 0.50–3             |
| Gross margin                            | 80–95%                 |
| Target blended CAC (verifier)           | INR 50,000–2,00,000    |
| Target LTV (12-month contract)          | INR 3,00,000–15,00,000 |
| LTV:CAC ratio (target)                  | > 3:1                  |

## 5.4 Revenue Milestones

Table 7: 12-month revenue roadmap

| Phase      | Milestone                                  | Target MRR |
|------------|--------------------------------------------|------------|
| Month 0–1  | Stellar Garage demo + 1 LOI                | INR 0      |
| Month 2–4  | First paid pilot (500–2,000 verifications) | INR 50K–2L |
| Month 5–8  | 3–5 verifier API clients                   | INR 2–5L   |
| Month 9–12 | Enterprise contract + 2 issuers            | INR 5–10L  |
| Year 2     | White-label + cross-border niche           | INR 15–30L |

## 6 Target Market & Use Cases

### 6.1 Primary Customer Segments

Table 8: Customer segments and willingness to pay

| Segment         | Who Pays               | Use Case                              | Budget Signal |
|-----------------|------------------------|---------------------------------------|---------------|
| <b>Verifier</b> | NBFC, fintech, BNPL    | Loan eligibility without full docs    | High          |
| <b>Verifier</b> | Crypto exchange        | KYC minimisation, on-ramp compliance  | High          |
| <b>Verifier</b> | PropTech, brokers      | Rental income verification            | Medium        |
| <b>Verifier</b> | DeFi protocol          | Collateral + income proofs on Stellar | Medium        |
| <b>Issuer</b>   | Employer, payroll SaaS | Issue income credentials once         | Medium        |
| <b>Issuer</b>   | Ed-tech, universities  | GPA/rank proofs for recruiters        | Medium        |

### 6.2 India-First Use Cases (Priority Order)

1. **Personal loan / BNPL underwriting** — Replace salary slip + full CIBIL with predicate proofs.
2. **Rental verification** — “Income  $\geq$  INR X” without payslip sharing.
3. **Crypto on-ramp KYC** — Exchanges verify compliance predicates; reduce PII retention liability.
4. **Campus placement** — “CGPA  $\geq$  7.0” without transcript upload to every recruiter.
5. **Cross-border remittance lending** — Diaspora income verification via Stellar-anchored credentials.

### 6.3 Total Addressable Market (India, Directional)

- India KYC/identity verification market: estimated **USD 1.5–2.5 B** (2025), growing 15–20% CAGR
- NBFC + fintech loan origination: **100M+** applications/year (directional)

- Addressable wedge: privacy-compliant predicate verification for regulated entities seeking DPDP alignment
- Initial SAM focus: 500 mid-size NBFCs/fintechs × INR 3–6 L annual verification spend = **INR 150–300 Cr** serviceable market entry point

## 7 Go-to-Market Strategy

### 7.1 Phase 1: Residency & Proof (July 2026)

- Deliver mainnet demo at Stellar Garage Final Pitching Day (25 July)
- Secure 2–3 LOIs from Rise In network intros (NBFC, fintech, or KYC partner)
- Publish open API documentation and public `/verify/:id` endpoint
- Launch “Verify with Krydo” QR flow for live audience demo

### 7.2 Phase 2: Paid Pilot (August–October 2026)

- Offer 90-day paid pilot: INR 50,000–2,00,000 for up to 2,000 verifications
- Mock issuer integration (employer/ed-tech sandbox credentials)
- Weekly metrics dashboard for pilot partner: verification volume, pass rate, latency
- Case study publication for sales collateral

### 7.3 Phase 3: Scale (Q4 2026–2027)

- Self-serve API signup for Starter tier verifiers
- Issuer partnership with one payroll or HR SaaS platform
- SCF grant application for Soroban on-chain verifier contract
- Compliance positioning: DPDP data minimisation, audit trail via on-chain anchors

### 7.4 Sales Motion

1. **Founder-led enterprise sales** for first 5 verifier clients
2. **Developer-first API** with sandbox keys and Postman collection
3. **Partner channel** via Rise In, Stellar ecosystem, and fintech accelerators
4. **Content marketing:** “Verification without over-collection” thought leadership

### 7.5 Key Messaging

#### Elevator Pitch

*“Krydo lets your customers prove they qualify—income, credit score, employment—without handing over their full financial life. You get a cryptographically verified yes/no. They keep their privacy. You reduce DPDP liability. Built on Stellar mainnet.”*

## 8 Stellar Garage Delhi: Program Plan

## 8.1 Program Overview

Table 9: Stellar Build Station Delhi — program parameters

| Parameter           | Detail                                |
|---------------------|---------------------------------------|
| Program             | Stellar Build Station Delhi           |
| Organisers          | Stellar & Rise In                     |
| Location            | Innov8 CP, Delhi NCR                  |
| Duration            | 21 days                               |
| Start date          | 4 July 2026                           |
| Demo Day            | 25 July 2026                          |
| Cohort size         | 15 selected teams                     |
| Format              | Hybrid with weekly in-person sessions |
| Build station hours | 11:00 AM – 4:30 PM                    |

## 8.2 In-Person Milestone Schedule

| Date                                                                                                                                            | Milestone                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| Objectives                                                                                                                                      |                                  |
| 4 July 2026                                                                                                                                     | Build Station Day 1<br>(Kickoff) |
| Cohort onboarding; workspace orientation; scope lock for Krydo on Stellar; Soroban project setup; initial testnet deploy; progress update #1.   |                                  |
| 11 July 2026                                                                                                                                    | Build Station Day 2              |
| Progress review with Rise In team; demo testnet happy-path (issue → prove → verify); incorporate feedback; confirm mainnet sponsorship process. |                                  |
| 18 July 2026                                                                                                                                    | Build Station Day 3              |

| Date                                                                                                                             | Milestone                       |
|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| Objectives                                                                                                                       |                                 |
| Pre-demo review;<br>go-to-market<br>strategy session;<br>pitch dry run (5–<br>7 min); mainnet<br>deploy checklist<br>finalised.  |                                 |
| 25 July 2026                                                                                                                     | Day 4 (Final Pitch-<br>ing Day) |
| Live main-<br>net demo<br>(mandatory);<br>final presen-<br>tations; prize<br>announcements;<br>ecosystem path-<br>way decisions. |                                 |

8.3 Builder Expectations & Compliance

Program Requirements — Non-Negotiable

- Maintain active progress throughout the 21-day period
- Attend all four in-person build station days (4, 11, 18, 25 July 2026)
- Submit regular progress updates with screenshots or demo recordings
- Deliver a functional, demo-ready product by program close
- **Deploy to Stellar mainnet before Final Pitching Day** (Rise In sponsors mainnet costs)
- Remain engaged, communicative, and collaborative within the cohort

*Note: Participants with sustained low engagement may be removed at organisers' discretion.*

## 8.4 21-Day Build Timeline

Table 11: Detailed build schedule

| Period         | Deliverables                                                                                                                                                 |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 Jul (Day 1)  | Kickoff pitch; Stellar scope lock; Soroban CLI + Rust setup; Freighter wallet; deploy KrydoAuthority on testnet; progress update #1.                         |
| 5–10 Jul       | Port KrydoCredentials + KrydoAudit to Soroban; Stellar SDK backend adapter; wallet connect (Freighter); one end-to-end testnet flow; weekly progress update. |
| 11 Jul (Day 2) | Rise In progress demo; feedback sprint; mainnet deploy plan; business LOI outreach via cohort network.                                                       |
| 12–17 Jul      | UI polish; error states; demo script; GTM one-pager; testnet regression; pitch deck v2.                                                                      |
| 18 Jul (Day 3) | Pitch dry run; GTM session; mainnet pre-deploy on staging; backup demo environment.                                                                          |
| 19–24 Jul      | <b>Mainnet deployment</b> ; live URL verification; final pitch rehearsal; documentation freeze.                                                              |
| 25 Jul (Day 4) | Live mainnet demo; Q&A; prize pitch; post-program pilot follow-ups.                                                                                          |

## 9 Technical Roadmap: Ethereum to Stellar

### 9.1 Migration Scope

The existing Ethereum Sepolia MVP provides production-grade business logic, API design, ZK engine, and frontend UX. The Stellar residency focuses on **contract port + wallet/auth adaptation + mainnet deployment**, not a full rewrite of the ZK cryptography layer.

Table 12: Component migration matrix

| Component                   | Current (Ethereum)          | Target (Stellar)                                      |
|-----------------------------|-----------------------------|-------------------------------------------------------|
| <b>Authority contract</b>   | KrydoAuthority.sol          | Soroban Rust: issuer whitelist                        |
| <b>Credentials contract</b> | KrydoCredentials.sol        | Soroban Rust: hash anchor + revocation                |
| <b>Audit contract</b>       | KrydoAudit.sol              | Soroban Rust: event anchoring                         |
| <b>Blockchain SDK</b>       | ethers v6                   | @stellar/stellar-sdk + Soroban RPC                    |
| <b>Wallet</b>               | MetaMask, wagmi, RainbowKit | Freighter, Albedo, Stellar Wallet Kit                 |
| <b>Auth</b>                 | EIP-4361 SIWE + JWT         | Stellar signature auth + JWT                          |
| <b>ZK engine</b>            | secp256k1 Pedersen + sigma  | Off-chain proofs retained; on-chain anchor only (MVP) |
| <b>Database</b>             | Firebase Firestore          | Unchanged                                             |
| <b>Frontend</b>             | React + Vite                | Adapt wallet layer; update network badges             |

## 9.2 MVP Scope Constraint (21 Days)

### Ruthless Scope for Demo Day

- **One credential type:** Income verification (range\_above proof)
- **One proof type:** range\_above (e.g. income  $\geq$  INR 10 Lakh)
- **Three Soroban contracts:** Authority, Credentials, Audit (minimal versions)
- **One happy-path demo:** Issue  $\rightarrow$  Generate proof  $\rightarrow$  Verify via public URL/QR
- **Mainnet live** with documented contract addresses and explorer links

## 9.3 Architecture on Stellar

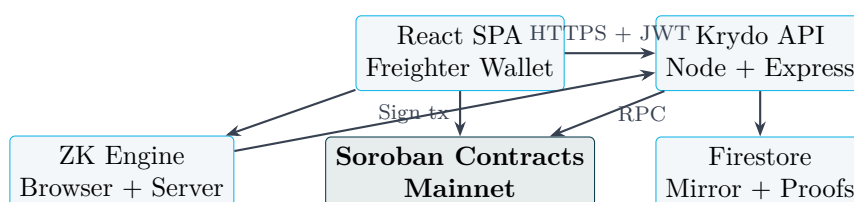


Figure 1: Krydo on Stellar — high-level architecture

## 9.4 Post-Residency Technical Debt

1. On-chain Soroban ZK verifier (Groth16/PLONK) for trustless verification
2. Ed25519-native Pedersen commitments (full cryptographic port from secp256k1)
3. Multi-sig root authority and decentralised issuer governance
4. Third-party cryptographic audit before regulated financial production use
5. SOC-2 equivalent backend certification

## 10 Demo Day Strategy (25 July 2026)

### 10.1 Three-Act Live Demo Script (5–7 Minutes)

1. **Act 1 — Trust Setup (60 sec):** Show whitelisted issuer on Stellar mainnet explorer. Explain Root Authority  $\rightarrow$  Issuer trust chain.
2. **Act 2 — Credential + Proof (2 min):** Issue income credential; user generates ZK proof in browser; show commitment math briefly (optional slide).
3. **Act 3 — Verification (2 min):** Lender scans QR / enters proof ID; API returns `valid: true, reason: "income  $\geq$  INR 10 Lakh"`; emphasise actual salary never transmitted.
4. **Close (60 sec):** Business model, India DPDP angle, pilot ask, Stellar cross-border vision.

### 10.2 Pitch Deck Structure (Recommended Slides)

1. Title + one-liner
2. Problem: over-collection in Indian verification
3. Solution: Krydo four-actor model + live demo screenshot
4. Why ZK (not fake privacy): sigma protocols, on-chain anchor

5. Why Stellar: cost, speed, cross-border, Rise In support
6. Business model: who pays, unit economics, INR worked example
7. Market: NBFC/fintech wedge, DPDP tailwind
8. Traction plan: residency demo, LOI targets, 90-day pilot
9. Team
10. Ask: pilot intros, mainnet sponsorship confirmation, SCF pathway

### 10.3 Backup Plan

- Pre-recorded demo video (2 min) if live network fails
- Testnet fallback environment with mainnet addresses shown separately
- Printed QR codes tested 24 hours before pitch

## 11 Financial Projections (Directional)

### 11.1 Year 1 P&L Summary (Conservative Scenario)

Table 13: Year 1 financial outlook (INR, conservative)

| Line Item                              | Year 1           | Notes                    |
|----------------------------------------|------------------|--------------------------|
| Revenue — pilot contracts              | 6,00,000         | 2–3 paid pilots          |
| Revenue — API subscriptions            | 18,00,000        | 4–6 verifier clients     |
| Revenue — issuer onboarding            | 4,00,000         | 1–2 issuers              |
| <b>Total Revenue</b>                   | <b>28,00,000</b> |                          |
| Infrastructure (Render, Firebase, RPC) | 1,20,000         |                          |
| Stellar mainnet transaction costs      | 30,000           | Subsidised early stage   |
| Team (founders + 1 contractor)         | 18,00,000        | Lean team                |
| Legal & compliance                     | 2,00,000         |                          |
| Sales & marketing                      | 1,50,000         |                          |
| <b>Total Costs</b>                     | <b>23,00,000</b> |                          |
| <b>Net (pre-grant)</b>                 | <b>5,00,000</b>  | SCF grant extends runway |

### 11.2 Funding Strategy

1. **Immediate:** Stellar Garage mainnet sponsorship (in-kind)
2. **Month 2–4:** SCF Build Award application (USD 50–150K range, directional)
3. **Month 6+:** Angel/pre-seed from fintech-focused angels (INR 50L–1.5Cr) post-pilot traction
4. **Avoid:** Raising before first paid pilot—LOI + pilot revenue de-risks valuation

## 12 Risk Analysis & Mitigation

Table 14: Risk register

| Risk                                       | Severity | Impact                    | Mitigation                                                        |
|--------------------------------------------|----------|---------------------------|-------------------------------------------------------------------|
| <b>ZK crypto port delays</b>               | High     | Mainnet demo scope creep  | Keep proofs off-chain; anchor hashes on Soroban only for MVP      |
| <b>21-day time-line insufficient</b>       | High     | Incomplete mainnet deploy | Ruthless scope cut; pre-built Ethereum demo as fallback narrative |
| <b>No paying customers post-residency</b>  | Medium   | Revenue gap               | LOI outreach during residency; 90-day paid pilot offer            |
| <b>Regulated entity sales cycle</b>        | Medium   | Slow revenue              | Start with crypto exchanges and PropTech (faster procurement)     |
| <b>Backend ZK verifier trust</b>           | Medium   | Enterprise objection      | Roadmap on-chain verifier; third-party audit plan                 |
| <b>Issuer adoption chicken-and-egg</b>     | Medium   | Empty credential network  | Sandbox/mock issuers for pilot; partner with one payroll SaaS     |
| <b>Mainnet deploy failure on pitch day</b> | Low      | Program disqualification  | Deploy by 23 July; 48-hour buffer; recorded backup demo           |

## 13 Competitive Moat & Defensibility

1. **Trust network ownership:** Root Authority controls issuer whitelist—credentials without Krydo approval are not trusted by the network.
2. **Cryptographic implementation:** 154 tests, sigma protocol engine, Pedersen commitments—not a wrapper around basic hashing.
3. **Dual-layer architecture:** On-chain truth (what exists, who issued) + off-chain performance (Firestore queries, proof storage).
4. **Regulatory positioning:** First-mover narrative on DPDP-aligned data minimisation for Indian fintech.
5. **Stellar ecosystem lock-in:** Early mainnet deployment + SCF relationship + cross-border use case development.
6. **API-first distribution:** Embedded verification becomes infrastructure; high switching costs once integrated into loan origination flow.

## 14 Conclusion & Immediate Next Steps

Krydo addresses a real, paid problem in India's verification infrastructure: **organisations need to know if someone qualifies; they do not need to own that person's entire financial identity.** Deploying on Stellar mainnet via the Stellar Garage Delhi residency provides the credibility, ecosystem access, and technical forcing function to convert a strong engineering MVP into a fundable B2B business.

## 14.1 Immediate Action Items

| # | Action                                           | Deadline       |
|---|--------------------------------------------------|----------------|
| 1 | Attend Build Station Day 1 kickoff at Innov8 CP  | 4 Jul 2026     |
| 2 | Complete Soroban dev environment setup           | 5 Jul 2026     |
| 3 | Deploy Authority contract to Stellar testnet     | 7 Jul 2026     |
| 4 | End-to-end testnet demo (issue → prove → verify) | 10 Jul 2026    |
| 5 | Progress update with screen recording            | Weekly         |
| 6 | Rise In mainnet sponsorship confirmation         | 11 Jul 2026    |
| 7 | Mainnet deployment                               | 22–23 Jul 2026 |
| 8 | Final pitch + live mainnet demo                  | 25 Jul 2026    |
| 9 | First paid pilot LOI signed                      | 31 Aug 2026    |

### Final Statement

*Prove you qualify — without revealing what you have.*  
 Krydo on Stellar Mainnet | Stellar Garage Delhi 2026

## A Appendix A: API Endpoints (Reference)

| Method | Endpoint               | Purpose                                |
|--------|------------------------|----------------------------------------|
| POST   | /api/zk/generate       | Generate ZK proof from credential      |
| POST   | /api/zk/verify         | Verify proof; return predicate result  |
| GET    | /verify/:id            | Public verification page (QR target)   |
| POST   | /api/credentials/issue | Issue credential (issuer role)         |
| GET    | /api/issuers           | List whitelisted issuers               |
| POST   | /api/auth/siwe         | Wallet authentication (Stellar adapt.) |

## B Appendix B: Progress Update Template

Week [N] Progress Update -- Krydo on Stellar

Completed:

- [Specific deliverable with screenshot/recording link]

Metrics:

- Testnet transactions: [N]
- API endpoints working: [list]
- Tests passing: [N/N]

Next 3 Days:

- [ ] Task 1
- [ ] Task 2

- [ ] Task 3

Blockers:

- [None / describe and ask for help]

## C Appendix C: Document & Resource Links

---

- Repository: <https://github.com/nishant-uxs/krydo>
- Stellar Documentation: <https://developers.stellar.org/>
- Soroban Smart Contracts: <https://soroban.stellar.org/>
- Stellar Community Fund: <https://communityfund.stellar.org/>
- Rise In: <https://risein.com/>
- Technical Documentation: DOCUMENTATION.md (project repository)
- Deployment Guide: DEPLOY.md (project repository)